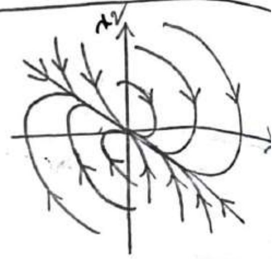
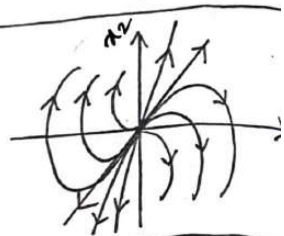
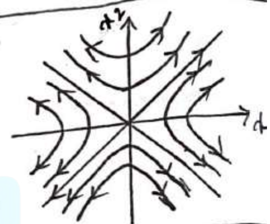
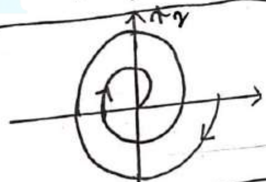
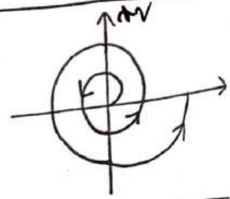
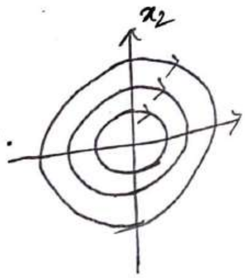

KTU
NOTES

KTU 2019 EEE S7

ADVANCED CONTROL SYSTEMS

MODULE 5

Phase Plane and Lyapunov Stability Analysis : Phase plots: Concepts- Singular points – Classification of singular points. Definition of stability- asymptotic stability and instability. Construction of phase trajectories using Isocline method for linear and nonlinear systems. Lyapunov stability analysis: Lyapunov function- Lyapunov methods to stability of nonlinear systems- Lyapunov methods to LTI continuous time systems.

Eigen values of system matrix	Type of singular point	Phase portrait of the system with singular point at origin
Distinct, real and the two eigen values are ~~positive~~ -ve	Stable	
Distinct, real and two eigen values are +ve	unstable node	
distinct, real, one eigen value is +ve and the other is -ve	saddle point	
complex conjugate with -ve real part.	stable focus	
complex conjugate with +ve real part.	unstable focus	
Purely imaginary and conjugate	centre or vortex point	

Phase Plane and Phase Trajectories

Phase plane method is a graphical method for the analysis of linear & nonlinear systems.

consider a 2nd order sys with equation,

$$\frac{d^2x}{dt^2} + 2\tau\omega_n\frac{dx}{dt} + \omega_n^2 x = 0$$

; τ – damping ratio

ω_n – natural frequency of oscillation

Let the state variables be x_1 & x_2

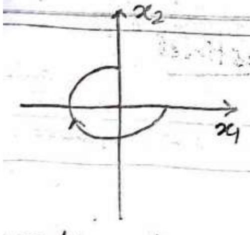
$$x_1 = x \qquad x_2 = \frac{dx}{dt}$$

$$\therefore \dot{x}_2 + 2\tau\omega_n x_2 + \omega_n^2 x_1 = 0$$

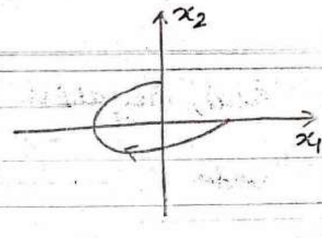
$$\left.\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\omega_n^2 x_1 - 2\tau\omega_n x_2 \end{aligned}\right\} \text{state equations}$$

The plane with state variable x_1, and x_2 as two axes is called phase plane. A trajectory can be constructed in phase plane between different values for x_1 & x_2 and is known as phase trajectory.

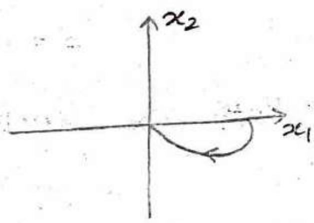
$\tau = 0$



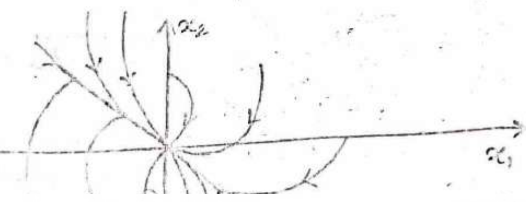
undamped



underdamped
$0 < \tau < 1$



overdamped
$\tau > 1$



damped $\tau = 1$

Singular Point

The phase plane method uses state variable approach. $x(t)$ and $\dot{x}(t)$ are the 2 state variables.

The point in the phase plane at which derivative of state variables are zero, called singular point.

(ie $\dot{x}_1 = 0$, & $\dot{x}_2 = 0$)

Nature of singular point is dependent on the roots of characteristic equation (or eigen values) of system. Depending on the position of roots in s plane, we get different kind of singularity.

Q. Determine the kind of singularity for the differential equation $\ddot{y} + 3\dot{y} + 2y = 0$

Ans: $$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0$$

$$s^2Y(s) + 3sY(s) + 2Y(s) = 0$$

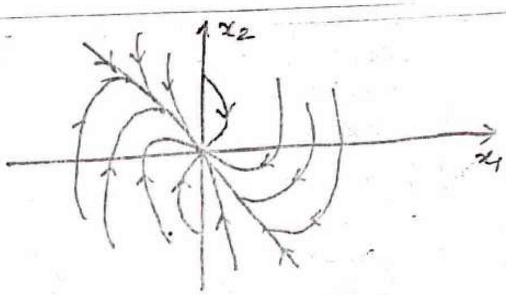
$$(s^2+3s+2)\,Y(s) = 0$$

$$s^2+3s+2 = 0$$

$$s = -1, -2$$

Both the roots are real, negative & distinct

$\therefore$ Stable node



A linear 2^{nd} order servo is described by the equation $\ddot{e} + 2\xi\omega_n\dot{e} + \omega_n^2 e = 0$

$\xi = 0.15$, $\omega_n = 1$ rad/sec. Find singular points

What kind of singularity is this.?

Ans: $\ddot{e} + 0.3\dot{e} + e = 0$

$e = x_1$

$\dot{e} = \dot{x}_1 = x_2$

$\ddot{e} = \dot{x}_2$

$\dot{x}_2 + 0.3x_2 + x_1 = 0$

$\dot{x}_2 = -0.3x_2 - x_1$

At singular point, $\dot{x}_1 = 0$

$\dot{x}_2 = 0$

$\dot{x}_1 = x_2 = 0$

$\dot{x}_2 = 0 \Rightarrow -0.3x_2 - x_1 = 0$

$x_1 = 0$

(0,0) is the singular point.

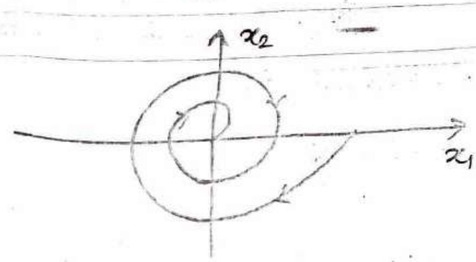
$\ddot{e} + 3\dot{e} + e = 0$

$s^2 e(s) + 3se(s) + e(s) = 0$

$s^2 + 3s + 1 = 0$

$s = \frac{-3+\sqrt{5}i}{2}, \frac{-3-\sqrt{5}i}{2}$

Both roots are complex conjugate with -ve real part. $\therefore$ Stable focus.



ISOCLINE METHOD

Procedure

- Draw the phase plane with x_1 on x axis & x_2 on Y ax
- Draw lines at different slopes : $[S = -2, -1, -0.5, 0, 1, 2]$
- Mark the initial point on x_2 axis, as given in question $(x_1(0), x_2(0))$, if not given assume same val on x_1 axis
- Phase trajectory leaves initial point at an angle
- Drop dotted lines at an angle $90°$ and at an angle $\theta = \tan^{-1}(s=2)$ (biggest slope in table) from initial point referred to x_2 axis.
- It meets the line $(s=2)$ at p and q. Half the len of pq gives the next point of the phase trajecto
- Phase trajectory leaves this point at an angle $\theta = \tan^{-1}(s=2)$
- Repeat the same procedure with $\theta = \tan^{-1}(s=2)$ an $\theta = \tan^{-1}(s=1)$ [2^{nd} last slope in the table]
- Join the points to obtain phase trajectory.

Q. A linear 2nd order servo is described by the equation $\ddot{e} + 2\delta\omega_n\dot{e} + \omega_n^2 e = 0$ where $\delta = 0.15$, $\omega_n = 1$ rad/sec, $e(0) = 1.5$ & $\dot{e}(0) = 0$.

Determine the singular points, construct the phase trajectory using the method of isoclines?

Ans: Let $x_1 = e$

$$x_2 = \dot{e}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\omega_n^2 x_1 - 2\delta\omega_n x_2$$

$$= -x_1 - 0.3x_2$$

$$\dot{x}_1 = 0 \implies x_2 = 0$$

$$\dot{x}_2 = 0 \implies -x_1 - 0.3x_2 = 0$$

$$x_1 = 0$$

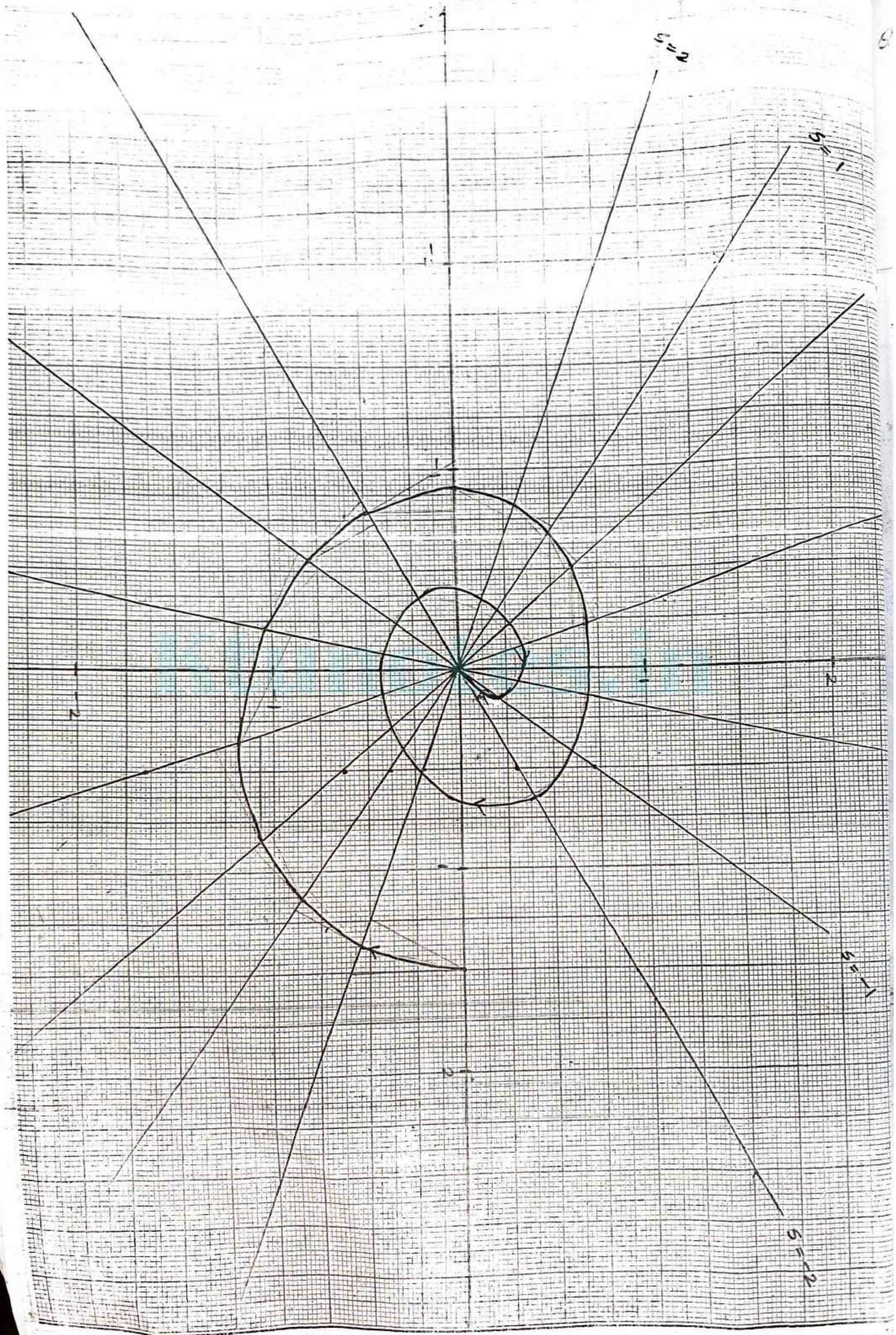
singular point $\Rightarrow (0,0)$

$$s = \frac{\dot{x}_2}{\dot{x}_1} = -\frac{(x_1 + 0.3x_2)}{x_2}$$

$$\boxed{x_2 = \frac{x_1}{-0.3 - s}}$$

$$\boxed{\alpha = \tan^{-1} s}$$

s	-2	-1	-0.5	0	0.5	1	2
α	-63°	-45°	-27°	0	27°	45°	63°
x_1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
x_2	0.294	0.714	2.5	-1.66	-0.625	-0.38	-0.21



8. Determine the stability of the system using Lyapunov's method.

$$\dot{x} = Ax, \quad A = \begin{bmatrix} -1 & 1 \\ -2 & -4 \end{bmatrix}$$

Ans: $\boxed{A^T P + PA = -Q}$ $\boxed{P_{12} = P_{21}}$

$$\begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}\begin{bmatrix} -1 & 1 \\ -2 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -P_{11} - 2P_{21} & -P_{12} - 2P_{22} \\ P_{11} - 4P_{21} & P_{12} - 4P_{22} \end{bmatrix} + \begin{bmatrix} -P_{11} - 2P_{12} & P_{11} - 4P_{12} \\ -P_{21} - 2P_{22} & P_{21} - 4P_{22} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -2P_{11} - 4P_{12} & P_{11} - 5P_{12} - 2P_{22} \\ P_{11} - 5P_{12} - 2P_{22} & 2P_{12} - 8P_{22} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$-2P_{11} - 4P_{12} \overset{+8P_{22}}{=} -1 \quad \text{--- (1)}$$

$$P_{11} - 5P_{12} - 2P_{22} = 0 \quad \text{--- (2)}$$

$$2P_{12} - 8P_{22} = -1 \quad \text{--- (3)}$$

Solving (1), (2) & (3) $P_{11} = 13/30$ $P_{12} = P_{21} = 1/30$, $P_{22} = 2/15$

$$\therefore P = \begin{bmatrix} 13/30 & 1/30 \\ 1/30 & 2/15 \end{bmatrix}$$

$13/30 > 0$, $|P| > 0$

$\therefore$ S/m is asymptotically stable.

9. $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ Ans: $\begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 1 \end{bmatrix}$

7. Investigate the stability of the nonlinear system using Lyapunov's second method:

$\dot{x}_1 = x_2$

$\dot{x}_2 = -x_1 - x_1^2 x_2$ $\qquad\qquad \frac{\partial V}{\partial t} = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2$

Ans: Let $V = x_1^2 + x_2^2$

$\dot{V} = 2x_1\dot{x}_1 + 2x_2\dot{x}_2$

$= 2x_1x_2 + 2x_2(-x_1 - x_1^2 x_2)$

$= 2x_1x_2 - 2x_1x_2 - 2x_1^2x_2^2$

$= \underline{-2x_1^2x_2^2}$

$V(x) > 0$ & derivative < 0 so sIm is asymptotic stable.

8. $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_2 - x_1^3$ if $V(x) = x_1^4 + x_1^2 + 2x_1x_2 +$

Ans: $V(x) = x_1^4 + x_1^2 + 2x_1x_2 + 2x_2^2$

$\dot{V}(x) = (4x_1^3 + 2x_1 + 2x_2)\dot{x}_1 + (2x_1 + 4x_2)\dot{x}_2$

$= 4x_1^3x_2 + 2x_1x_2 + 2x_2^2 + (2x_1 + 4x_2)(-x_2 - x_1^3)$

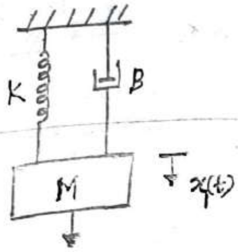
$= 4x_1^3x_2 + 2x_1x_2 + 2x_2^2 - 2x_1x_2 - 2x_1^4 - 4x_2^2 - 4x$

$= -2x_2^2 - 2x_1^4$

Asymptotically stable

Lyapunov Stability

Consider a spring mass damper system,



The equation is,

$$M\frac{d^2x_1}{dt^2} + B\frac{dx_1}{dt} + Kx_1 = 0$$

$$M\frac{d^2x_1}{dt^2} = -B\frac{dx_1}{dt} - Kx_1 \qquad \text{Let } M=1$$

$$\frac{d^2x_1}{dt^2} = -B\frac{dx_1}{dt} - Kx_1$$

$$\ddot{x}_1 = -B\dot{x}_1 - Kx_1$$

Let x_1 & x_2 be two state variables,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \ddot{x}_1 = -Kx_1 - Bx_2$$

This method is based on the concept of energy.

At any instant, the total energy V of the s/m consists of Kinetic energy of mass M and potential energy stored in spring.

$$\therefore V(x_1, x_2) = \tfrac{1}{2} M x_2^2 + \tfrac{1}{2} K x_1^2 \qquad (M=1)$$

for $x>0$, & $x \neq 0$, $V(x) > 0$ & $V(0) > 0$

It indicates that total energy is positive

Rate of change of energy,

$$\frac{dV(x_1, x_2)}{dt} = \frac{dV}{dx_1}\cdot\frac{dx_1}{dt} + \frac{dV}{dx_2}\cdot\frac{dx_2}{dt}$$

$$\frac{\partial V}{\partial x_1} = \tfrac{1}{2} K\, 2x_1 = Kx_1$$

$$\frac{\partial V}{\partial x_2} = \tfrac{1}{2}\times 2x_2 = x_2$$

$$\frac{dx_1}{dt} = \dot{x}_1 = x_2 \qquad \frac{dx_2}{dt} = \dot{x}_2 = -Kx_1 - Bx_2$$

$\frac{dV}{dt}$ is $-ve$ at all point except at $x_2 = 0$

Lyapnov's method in its simplest form is given by following theorems

Theorem 1

$\dot{x}_1 = f(x)$; $f(0) = 0$

Suppose there exists a scalar function $V(x)$, for some real number: $t > 0$, satisfies the following properties for all x in the region $|x| \leq t$

(a) $V(x) > 0$; $x \neq 0$

(b) $V(0) = 0$

(c) $V(x)$ has non-linear partial derivatives

(d) $\frac{dV}{dt} \leq 0$ (ie $\frac{dV}{dt}$ is negative semi definite)

Then the system is stable at the origin.

Theorem 2

If the property (d) of theorem 1 is replaced with $\frac{dV}{dt} < 0$, $x \neq 0$ (ie $\frac{dV}{dt}$ is negative definite) then the system is asymptotically stable

Theorem 3

If all the conditions of theorem 2 hold and in addition, $V(x) \to \infty$ as $|x| \to \infty$ then the s/m is asymptotically stable in the

Theorem 4

consider the s/m $\dot{x} = f(x)$, $f(0) = 0$

Suppose there exists a scalar function $w(x)$ which for some real number $t > 0$, satisfies the following properties for all x in the region,

$$|x| \leq t$$

(a) $w(x) > 0$; $x \neq 0$

(b) $w(0) = 0$

(c) $w(x)$ has continuous partial derivatives with all components of x.

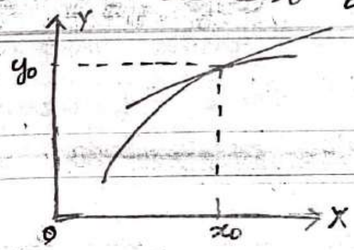
(d) $\frac{dv}{dt} \geq 0$

Then this s/m is unstable at origin.

Linearisation of Non-linear System

Practically none of the physical s/m in nature are perfectly linear. So for the ease of calculations, non-linear s/m can be approximated to linear system.

consider general element with input $x(t)$ & o/p $y(t)$ as in fig,



Expansion of the equation $y = f(x)$ into a Taylor's series about the normal operating point (x_0, y_0) gives,

$$y = f(x) = f(x_0) + \left.\frac{df}{dx}\right|_{x=x_0} \frac{(x-x_0)}{1!} + \left.\frac{d^2f}{dx^2}\right|_{x=x_0} \frac{(x-x_0)^2}{2!} + \cdots$$

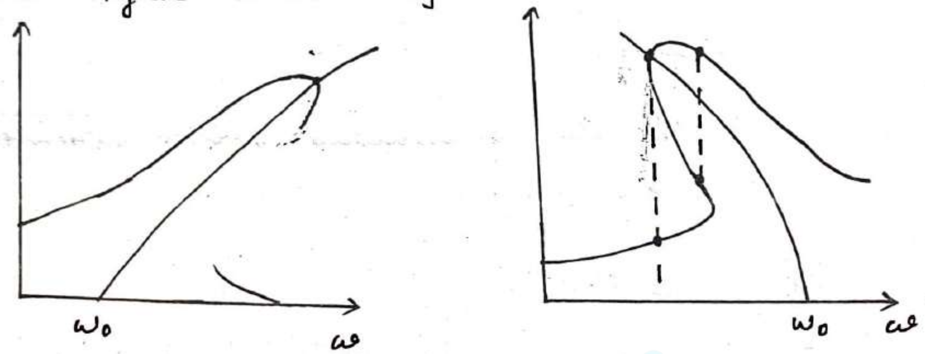
By neglecting higher order terms, $y = f(x_0) + \frac{df}{dx}(x - x_0)$

So the non-linear ... approximate ... linear system.

Characteristics of Non-linear s/m

① Jump Resonance

In the frequency response of non-linear s/ms the amplitude of o/p may jump from one point to another by varying the value of ω. This phenomenon is called jump resonance. It is shown in the figure when subjected to sinusoidal input.



② Limit Cycles

The output of non-linear s/m may exhibit oscillations with fixed amplitude and frequency. These oscillations are limit cycle.

③ Subharmonic Oscillation

When a non-linear system is excited by sinusoidal signal, the output will have steady state oscillations whose frequency is an integral submultiple of force frequency. These oscillations are called subharmonic oscillations.

④ Backlash

non-linearity occuring in physical s/m is hysteresis in mechanical transmissions (gear trains & linkages) is commonly referred as backlash. When the input is increased o/p may increased in one p... and ... o/p may decreased

Advanced Control Systems

Module V

Part 1 : Phase Plane Analysis

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Syllabus

5	**Phase Plane and Lyapunov Stability Analysis**	**(8 hours)**
5.1	Phase plots: Concepts- Singular points - Classification of singular points.	1
5.2	Construction of phase trajectories using Isocline method for linear and nonlinear systems	2
5.3	Definition of stability- asymptotic stability and instability	1
5.4	Lyapunov stability analysis: Lyapunov function- Lyapunov methods to stability of nonlinear systems	2
5.5	Lyapunov methods to LTI continuous time systems.	2

Phase plane Analysis

- **Phase Plane Analysis is a graphical method for studying second-order systems respect to initial conditions by:**
  - **Providing motion trajectories corresponding to various initial conditions.**
  - **Examining the qualitative features of the trajectories**
  - **Obtaining information regarding the stability of the equilibrium points**

Advantages of Phase Plane Analysis

- It is graphical analysis and the solution trajectories can be represented by curves in a plane
- Provides easy visualization of the system qualitative
- Without solving the nonlinear equations analytically, one can study the behavior of the nonlinear system from various initial conditions.
- It is not restricted to small or smooth nonlinearities and applies equally well to strong and hard nonlinearities.
- There are lots of practical systems which can be approximated by second-order systems, and apply phase plane analysis.

Disadvantages of Phase Plane Method

- **It is restricted to at most second-order**
- **Graphical study of higher-order is computationally and geometrically complex.**

Phase Trajectories

The curve describing the state point (x_1, x_2) in the phase-plane with time as running parameter is called phase trajectory [i.e., the locus of state point (x_1, x_2) in phase plane is called phase trajectory]. A trajectory can be constructed in phase-plane for each set of initial conditions. Hence a family of trajectories can be constructed for a system in a phase plane and such a family of trajectories is called a phase portrait. Typical phase trajectories of a second order system for different values of damping is shown in fig 2.40 and the phase portrait of the system for critical damping is shown in fig 2.41.

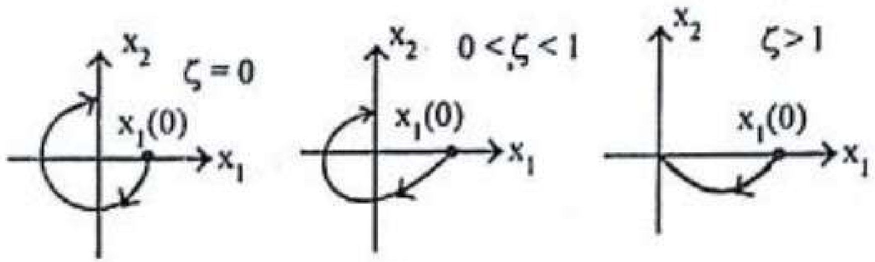


Fig 2.40 : Phase trajectories of a second order system for different values of ζ and with initial condition $x_1(0)$.

Singular Point

Singular Point is also called Equilibrium point

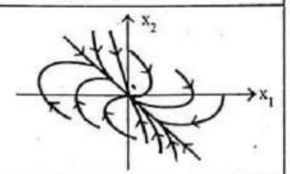
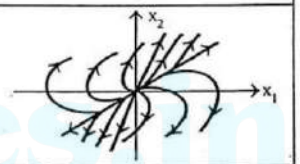
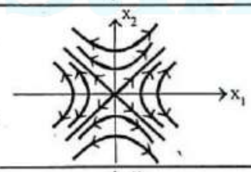
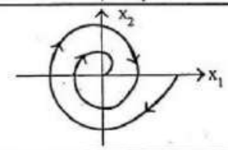
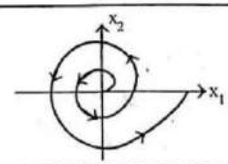
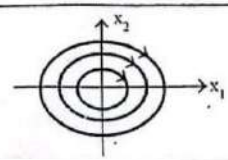
SINGULAR POINTS

A point in phase-plane at which the derivative of all state variables are zero is called singular point. It is also called equilibrium point. If the system is placed at a singular or equilibrium point, it will continue to lie there if left undisturbed (i.e., the derivatives of all the phase variables are zero and so the system state remains unchanged).

The singular points are classified as Nodal point, Saddle point, Focus point and Centre or Vortex point depending on the eigen values of the system matrix, A. The table 2.1 shows the phase portrait of systems with various types of singular points.

Types of Singular points

TABLE 2.1

Eigen Values of system matrix	Type of singular point	Phase portrait of the system with singular point at origin
Distinct, real and the two eigen values are negative	Stable node	
Distinct, real and the two eigen values are positive	Unstable node	
Distinct, real, one eigen value is positive and the other is negative	Saddle point	
Complex-conjugate with negative real part	Stable focus	
Complex conjugate with positive real part	Unstable focus	
Purely imaginary and conjugate	Centre or Vortex point	

Plotting Phase Plane Diagram

- Analytical Method
- Isocline Method
- Delta Method

1. Analytical Method

The state equations of a second order autonomous system are,

$$\dot{x}_1 = f_1(x_1, x_2) \qquad .....(2.120)$$

$$\dot{x}_2 = f_2(x_1, x_2) \qquad .....(2.121)$$

where x_1 and x_2 are the state variables of the system.

On dividing equ(2.121) by equ(2.120) we get,

$$\frac{\dot{x}_2}{\dot{x}_1} = \frac{f_2(x_1, x_2)}{f_1(x_1, x_2)} \qquad .....(2.122)$$

since $\dot{x}_2 = dx_2 / dt$ and $\dot{x}_1 = dx_1 / dt$, the equ (2.122) can be written as,

$$\frac{dx_2}{dx_1} = \frac{f_2(x_1, x_2)}{f_1(x_1, x_2)} \qquad .....(2.123)$$

2. Isocline Method

Let, S = Slope at any point in the phase-plane.

From equ(2.123) we get,

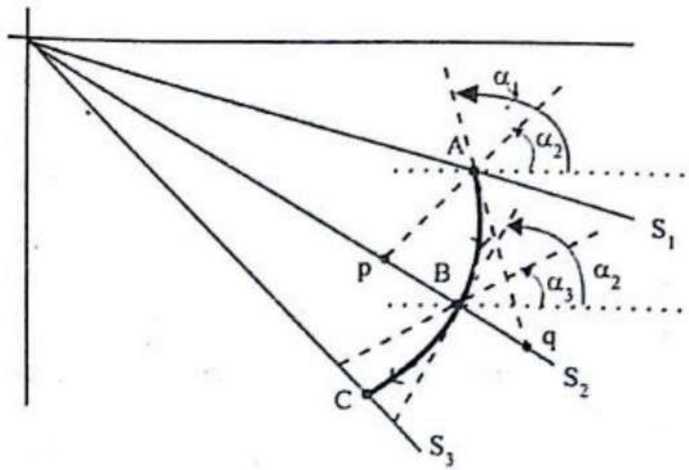
$$S = \frac{dx_2}{dx_1} = \frac{f_2(x_1, x_2)}{f_1(x_1, x_2)} \qquad .....(2.124)$$

Let, S_1 = Slope at a point on phase trajectory-1.

From equ(2.124) we get,

$$f_2(x_1, x_2) = S_1 \times f_1(x_1, x_2) \qquad .....(2.125)$$

The equ(2.125) defines the locus of all such points in phase-plane at which the slope of the phase-trajectory is S_1. A locus passing through the points of same slope in



$\alpha_1 = \tan^{-1}(S_1)$
$\alpha_2 = \tan^{-1}(S_2)$
$\alpha_3 = \tan^{-1}(S_3)$

Fig 2.44 : Construction of phase trajectory by isocline method

3 Delta Method

s

CONSTRUCTION OF PHASE TRAJECTORIES BY DELTA METHOD

Consider a second order linear or nonlinear system represented by the equation,

$$\ddot{x} + f(x, \dot{x}, t) = 0 \qquad .....(2.126)$$

The equation (2.126) can be converted to the form shown below,

$$\ddot{x} + \omega_n^2 [x + \delta(x, \dot{x}, t)] = 0 \qquad .....(2.127)$$

In equation (2.127), δ is a function of x, $\dot{x}$ and t, but for short intervals, the changes in these variables are negligible. Hence for a short interval, δ is considered as a constant.

$$\therefore \ddot{x} + \omega_n^2 (x + \delta) = 0 \qquad .....(2.128)$$

Delta Method

Let us choose the state variables x_1 and x_2 as,

$$x_1 = x \qquad .....(2.129)$$

$$x_2 = \dot{x} / \omega_n \qquad .....(2.130)$$

On differentiating equ(2.129) we get, $\dot{x}_1 = \dot{x}$

But $\dot{x} = \omega_n x_2$, $\quad \therefore \dot{x}_1 = \omega_n x_2 \qquad .....(2.131)$

On differentiating equ(2.130) we get,

$$\dot{x}_2 = \ddot{x} / \omega_n \quad \text{(or)} \quad \ddot{x} = \omega_n \dot{x}_2$$

On substituting $\ddot{x} = \omega_n \dot{x}_2$ and $x = x_1$ in equ(2.128) we get,

$$\omega_n \dot{x}_2 + \omega_n^2 (x_1 + \delta) = 0$$

$$\therefore \dot{x}_2 = -\omega_n (x_1 + \delta) \qquad .....(2.132)$$

The state equations are

$$\dot{x}_1 = \omega_n x_2$$

$$\dot{x}_2 = -\omega_n (x_1 + \delta)$$

From the state equations the slope equation over short interval can be written as,

$$\frac{\dot{x}_2}{\dot{x}_1} = \frac{dx_2}{dx_1} = \frac{-(x_1 + \delta)}{x_2} \qquad .....(2.133)$$

Delta Method

Let point A be a point on phase trajectory with coordinates (x_1, x_2) as shown in fig 2.45 a and b, (usually the point A will be the starting point of the trajectory obtained from the initial conditions).

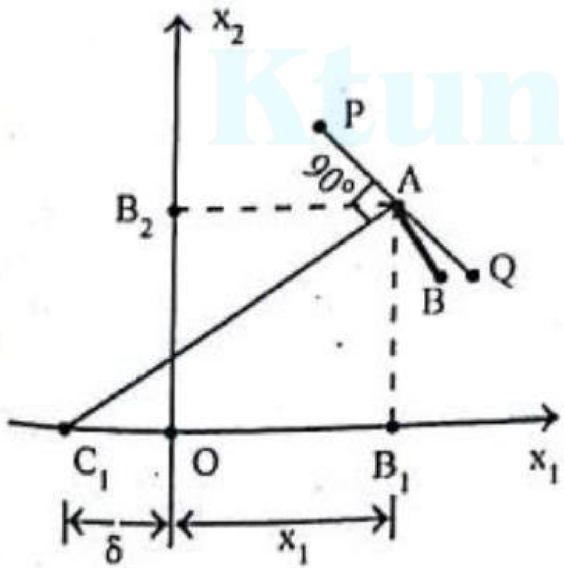


Fig : 2.45 a

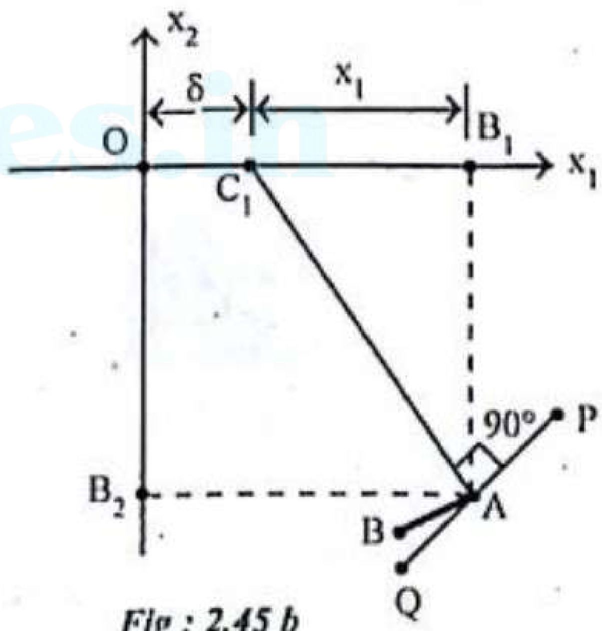


Fig : 2.45 b

Delta Method

Procedure to draw phase trajectory by delta method

1. The phase trajectory starts at point A, corresponding to initial conditions, $[x_1(0), x_2(0)]$. Calculate δ_1 using the initial conditions, (i.e., using the coordinates of point A). (Refer fig 2.6.1. of example 2.6.).
2. The coordinates of point C_1 is $(-\delta_1, 0)$ if δ_1 is positive or $(|\delta_1|, 0)$ if δ_1 is negative. Mark the point C_1 on x_1 axis. With point C_1 as centre and C_1A as radius draw a circular arc AB. The point B (new point)is an arbitrary point on phase plane. Accurate phase trajectory can be constructed if we draw very short segment to locate a new point.

   Note : A compass can be used to draw the circular arc segments.
3. Using the coordinates of point B, calculate δ_2. The coordinates of point C_2 is $(-\delta_2, 0)$ if δ_2 is positive or $(|\delta_2|, 0)$ if δ_2 is negative. Mark the point C_2 on x_1 axis. With C_2 as centre and C_2B as radius, draw a circular arc BC, to locate another new point, point-C.
4. Similar procedure is repeated at point C to locate point D and so on. The circular arc segments through the points A, B, C, D, etc., gives the phase trajectory.

The delta method results in time saving when a single or a few phase trajectories are required. Using delta method, the phase trajectory for any system with step or ramp input can be conveniently drawn.

Problem

1

A linear second order servo is described by the equation

$$\ddot{e} + 2\zeta\omega_n\dot{e} + \omega_n^2 e = 0$$

where $\zeta = 0.15$, $\omega_n = 1$ rad/sec, $e(0) = 1.5$ and $\dot{e}(0) = 0$.

Determine the singular point. Construct the phase trajectory, using the method of isoclines.

SOLUTION

Let x_1 and x_2 be the state variables of the system and they are related to the system variable, e as shown below.

$$x_1 = e \qquad .....(2.5.1)$$

$$x_2 = \dot{e} \qquad .....(2.5.2)$$

On differentiating equ(2.5.1) we get,

$$\dot{x}_1 = \dot{e}_1 = x_2 \qquad .....(2.5.3)$$

On differentiating equ(2.5.2) we get,

$$\dot{x}_2 = \ddot{e} \qquad .....(2.5.4)$$

Given that, $\ddot{e} + 2\zeta\omega_n\dot{e} + \omega_n^2 e = 0$ .....(2.5.5)

On substituting equation (2.5.1), (2.5.2) and (2.5.4) in equ (2.5.5) we get,

$$\dot{x}_2 + 2\zeta\omega_n x_2 + \omega_n^2 x_1 = 0$$

$$\therefore \dot{x}_2 = -\omega_n^2 x_1 - 2\zeta\omega_n x_2 \qquad .....(2.5.6)$$

Problem

2

The state equations of the system are given by equations (2.5.3) and (2.5.6)

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\omega_n^2 x_1 - 2\zeta\omega_n x_2$$

The singular point is obtained from state equations by putting $\dot{x}_1 = 0$ and $\dot{x}_2 = 0$

Let the coordinates of singular point in phase plane $= \left(x_1^0, x_2^0\right)$

On substituting $\dot{x}_1 = 0$ and $x_2 = x_2^0$ in equ(2.5.3) we get, $x_2^0 = 0$.

On substituting $\dot{x}_2 = 0$, $x_1 = x_1^0$ and $x_2 = x_2^0$ in equ(2.5.6) we get,

$$0 = -\omega_n^2 x_1^0 - 2\zeta\omega_n^2 x_2^0$$

But $x_2^0 = 0$, $\quad \therefore 0 = -\omega_n^2 x_1^0 \quad$ (or) $\quad x_1^0 = 0$

Therefore, the coordinates of singular point are $(0, 0)$ and so the origin is the singular point.

Problem

- 3

The slope of the phase trajectory is given by

$$S = \frac{dx_2/dt}{dx_1/dt} = \frac{\dot{x}_2}{\dot{x}_1} \qquad .....(2.5.7)$$

On substituting for $\dot{x}_1$ and $\dot{x}_2$ from equations (2.5.3) and (2.5.6) in equ(2.5.7) we get,

$$S = -\frac{(\omega_n^2 x_1 + 2\zeta\omega_n x_2)}{x_2} \qquad .....(2.5.8)$$

Put $\zeta = 0.15$ and $\omega_n = 1$, in equ(2.5.8)

$$S = \frac{-(x_1 + 2 \times 0.15\, x_2)}{x_2} = \frac{-(x_1 + 0.3\, x_2)}{x_2}$$

$$= \frac{-x_1}{x_2} - \frac{0.3x_2}{x_2} = \frac{-x_1}{x_2} - 0.3$$

$$\therefore \frac{x_1}{x_2} = -0.3 - S \qquad \text{(or)} \qquad \frac{x_2}{x_1} = \frac{1}{-0.3 - S}$$

$$\therefore x_2 = \frac{x_1}{-0.3 - S} \qquad .....(2.5.9)$$

Problem

- 4

From equ(2.5.9) we can conclude that the isoclines are straight lines. For each value of S we can draw one isocline.

Using equation (2.5.9), the coordinates (x_1, x_2) in the phase plane for various slopes can be calculated. Since there are three variables. Let us assume two variables and calculate the third variable.

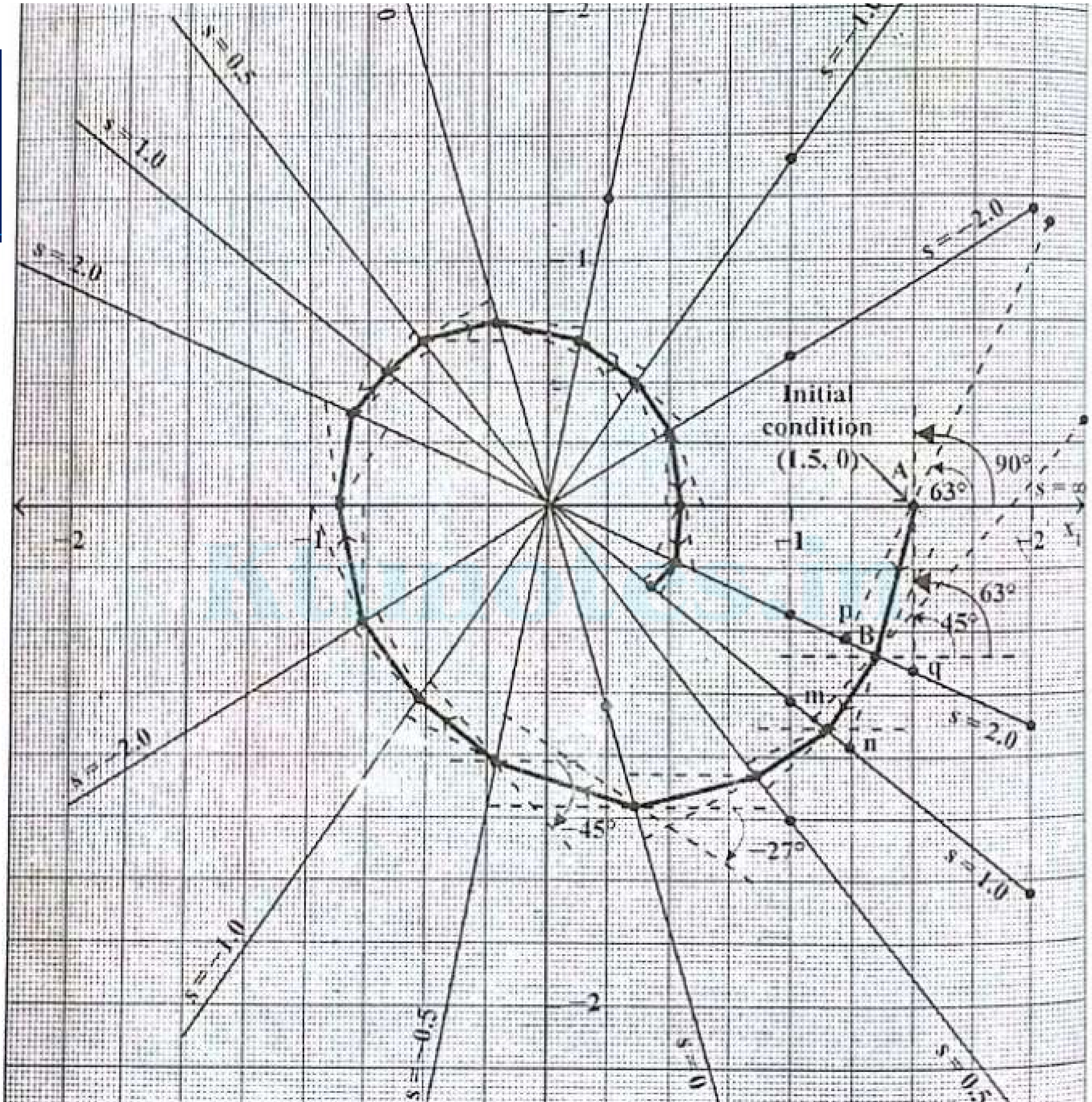
Let us choose values of S as $-2, -1.0, -0.5, 0, 0.5, 1.0$ and 2.0

For each value of S, choose two values of x_1 and calculate x_2 using equation (2.5.9). The values of S, x_1 and x_2 are tabulated. The slope angle, α is calculated for each value of S, using the expression, $\alpha = \tan^{-1}(S)$ and tabulated in table-2.5.1.

TABLE-2.5.1.

S	-2.0		-1.0		-0.5		0		0.5		1.0		2.0	
α	-63°		-45°		-27°		0		27°		45°		63°	
	x_1	x_2	x_1	x_2	x_1	x_2	x_1	x_2	x_1	x_2	x_1	x_2	x_1	x_2
	1.0	0.6	1.0	1.4	0.25	1.25	0.25	-0.8	1.0	-1.25	1.0	-0.77	1.0	-0.43
	2.0	1.2	1.5	2.1	0.5	2.5	0.75	-2.5	2.0	-2.5	2.0	-1.54	2.0	-0.86

• 5



Direction of positive and negative slope angle

Problem

- 6

The isoclines corresponding to each slope is drawn using the coordinates listed in table-2.5.1. and they are shown in fig 2.5.1.

The phase trajectory starts at point A on x_1 axis, (i.e., given initial condition is (1.5, 0)). The phase trajectory is shown in fig 2.5.1 and the construction of the trajectory is explained below.

The slope of the trajectory when it crosses x_1 axis is infinite. Hence draw a line at an angle, $\tan^{-1}\alpha = 90°$ with respect to x_1 axis and passing through point A. Let this line meet the isocline corresponding to S = 2.0 at point q. Then draw a line at an angle $\tan^{-1} 2.0 = 63°$ with respect to x_1 axis and passing through point A. Let this line meet the isocline corresponding to S = 2.0 at point p. Now the phase trajectory will pass through point B on the isocline corresponding to S_2. The point B lies at the middle of the segment pq. Draw a smooth curve between A and B, which is a section of phase trajectory.

Problem

- 7

In general, if F is the crossing point of phase trajectory with isocline-a corresponding to a slope of S_a and if the next isocline is isocline-b corresponding to a slope of S_b. Then draw two lines through point F, one at an angle of $\tan^{-1} S_a$ and the other at $\tan^{-1} S_b$. [The angles are marked from point F with respect to a (horizontal) line parallel to x_1 axis. Positive angles are measured in anticlockwise direction and negative angles are measured in clockwise direction]. These two lines will meet the isocline-b at points d and e. Now the crossing point of phase trajectory is fixed at the middle of d and e on the isocline-b.

Thus crossing point of phase trajectory on each isocline is determined. The complete phase trajectory is obtained by drawing a smooth curve through all the crossing points, as shown in fig 2.5.1.

RESULT

1. The singular point lies at origin.
2. From fig 2.5.1, it is observed that the phase trajectory spiral towards the origin, hence the type of singular point is stable focus.

Problem

-

EXAMPLE 2.6

Construct a phase trajectory by delta method for a nonlinear system represented by the differential equation, $\ddot{x} + 4|\dot{x}|\dot{x} + 4x = 0$. Choose the initial conditions as $x(0) = 1.0$ and $\dot{x}(0) = 0$.

SOLUTION

Given that, $\ddot{x} + 4|\dot{x}|\dot{x} + 4x = 0$ .....(2.6.1)

To construct phase trajectory by delta method, the equ(2.6.1) can be converted to the form,

$$\ddot{x} + \omega_n^2 (x + \delta) = 0 \qquad .....(2.6.2)$$

Hence, $\ddot{x} + 4|\dot{x}|\dot{x} + 4x = \ddot{x} + 4[x + |\dot{x}|\dot{x}] = 0$ .....(2.6.3)

On comparing equ(2.6.2) and (2.6.3) we get,

$$\delta = |\dot{x}|\dot{x} \qquad .....(2.6.4)$$

Problem

- 9

and $\omega_n^2 = 4, \quad \therefore \omega_n = 2$ rad / sec

Let us choose the state variables x_1 and x_2 such that,

$$x_1 = x \qquad \text{.....(2.6.5)}$$

$$\text{and} \quad x_2 = \frac{\dot{x}}{\omega_n} = \frac{\dot{x}}{2} \qquad \text{.....(2.6.6)}$$

From equ(2.6.6) we get, $\dot{x} = 2x_2$. On substituting $\dot{x} = 2x_2$ in equ(2.6.4) we get,

$$\delta = |\dot{x}|\dot{x} = |2x_2|\, 2x_2 = 4|x_2|x_2 \qquad \text{.....(2.6.7)}$$

The given initial conditions are $x_1(0) = 1.0$ and $x_2(0) = 0$.

Let point A be the starting point of the phase trajectory.

$\therefore$ Coordinates of point $A = (x_1(0), x_2(0)) = (1.0, 0)$

Let δ_1 = Value of delta at point A.

Since, $x_2(0) = 0, \delta_1 = 0$

Let C_1 = Centre of the circular arc AB.

Problem

- 10

The point C_1 is marked at $-\delta_1$ on x_1-axis. Since $\delta_1 = 0$, the point C_1 lies at origin.

With C_1 as centre and C_1A as radius, draw a circular arc AB as shown in fig 2.6.1. (Draw the circular arc using compass). The segment AB is a section of phase trajectory.

From the graph the value of x_2, corresponding to point B is $x_2 = -0.1875$. (Here the value of x_2 alone is needed to calculate δ).

The above procedure is repeated at point B to determine the segment BC of phase trajectory shown in fig 2.6.1. Similarly the segment CD, DE, EF, FG, GH, HI, IJ, JK, and KL shown in fig 2.6.1 are determined. The values of x_2, δ and centre of the circular arc segments (C_1, C_2, C_3,.........) are tabulated in table-2.6.1.

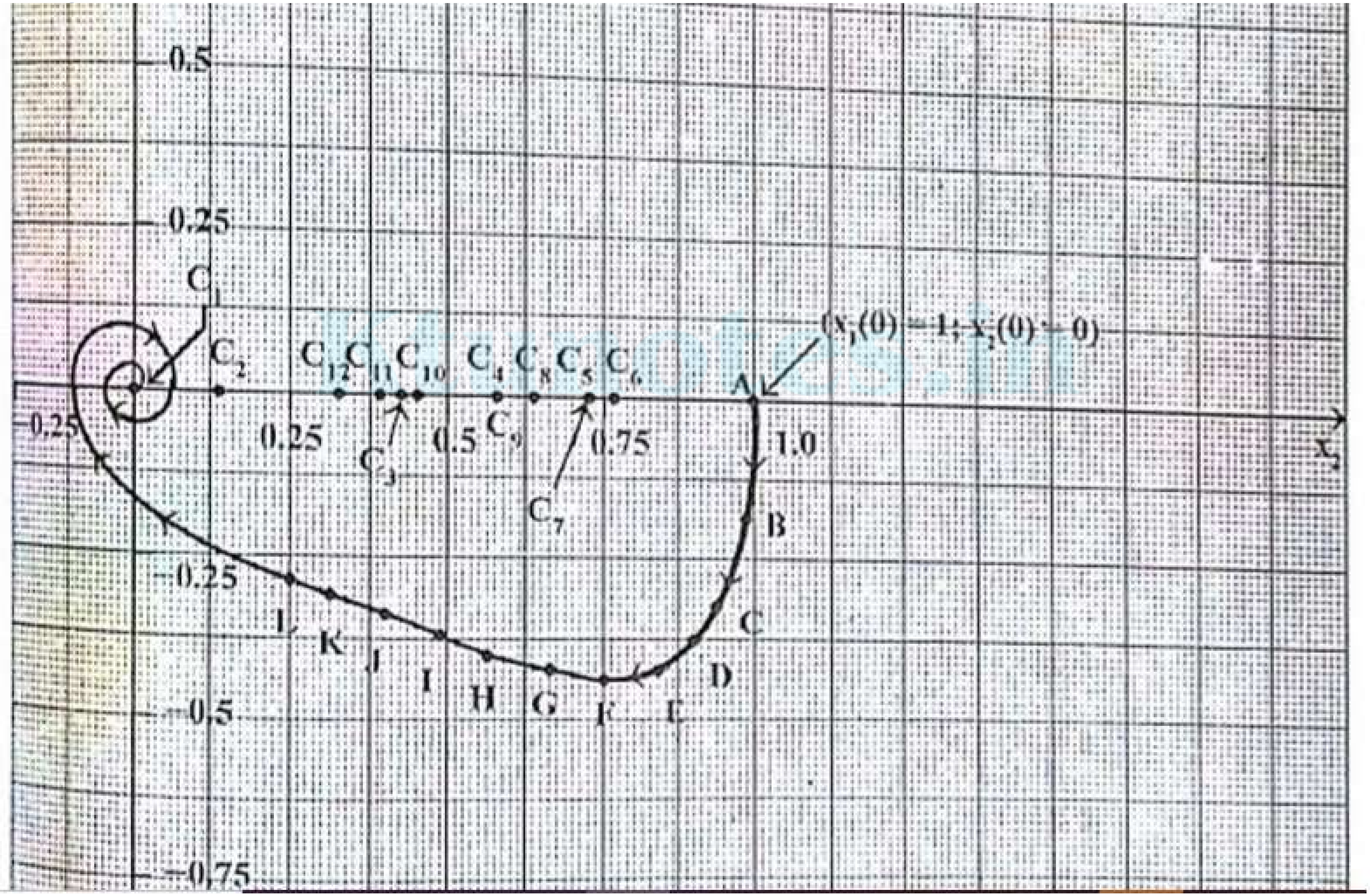
The complete phase trajectory shown in fig 2.6.1. is determined by repeating the above procedure again and again.

Problem

- 11

x_2		δ		$-\delta$	
0	(A)	0	(δ_1)	0	(C_1)
−0.1875	(B)	−0.140625	(δ_2)	0.140625	(C_2)
−0.325	(C)	−0.4225	(δ_3)	0.4225	(C_3)
−0.375	(D)	−0.5625	(δ_4)	0.5625	(C_4)
−0.425	(E)	−0.7225	(δ_5)	0.7225	(C_5)
−0.4375	(F)	−0.7656	(δ_6)	0.7656	(C_6)
−0.425	(G)	−0.7225	(δ_7)	0.7225	(C_7)
−0.4	(H)	−0.64	(δ_8)	0.64	(C_8)
−0.375	(I)	−0.5625	(δ_9)	0.5625	(C_9)
−0.3375	(J)	−0.4556	(δ_{10})	0.4556	(C_{10})
−0.3125	(K)	−0.3906	(δ_{11})	0.3906	(C_{11})
−0.2875	(L)	−0.3306	(δ_{12})	0.3306	(C_{12})

Problem

- 
0.5
0.25
C_1
C_2
C_{12} C_{11} C_{10} C_4 C_8 C_5 C_6
$(x_1(0) = 1;\ x_2(0) = 0)$
A
-0.25
0.25
C_3
0.5
C_9
0.75
C_7
1.0
x_1
B
C
D
E
F
G
H
I
J
K
L
-0.25
-0.5
-0.75

University Questions

- B

What are singular point? Explain the types of singular point.

Determine given quadratic form is positive definite or not

$$V(x) = 10x_1^2 + 4x_2^2 + x_3^2 + 2x_1 x_2 - 2x_2 x_3 - 4x_1 x_3$$

(10)

A second order system is represented by $\dot{x} = Ax$ where

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Assuming matrix Q to be identity matrix, solve for matrix P in the equation $A^T P + PA = -Q$. Use Lyapunov theorem and determine the stability of the system. Write the Lyapunov function $V(x)$

University Questions

- 2

Define Singular point. Explain the nature of Eigen values of system matrix for any five types of singular points.

Explain Liapunov second method of stability for nonlinear systems.

A linear second order system is described by the equation

$\ddot{e} + 2\delta\omega_n\dot{e} + \omega_n^2 e = 0$

Where $\delta = 0.15$, $\omega_n = 1$rad/sec, e(0)=1.5, and $\dot{e}(0) = 0$

Determine the singular point and state the stability by constructing the phase trajectory using the method of isoclines.

University Questions

- 3

Explain with neat diagram, what is phase trajectory and phase portrait?
Define positive definite and positive semi definite functions according to Liapunov stability criteria, with suitable examples.

16 Consider the following non linear differential equation. (10)

$$\ddot{y} - \left(0.1 - \frac{10}{3}\dot{y}^2\right)\dot{y} + y + y^2 = 0$$

Find all singular points of the system, classify them and sketch the phase portrait in the neighbourhood of singular points.

17 a) Discuss any three non linearities present in nature. (6)

b) Investigate the stability of the following non-linear system using Liapunov direct method (4)

$\dot{x}_1 = x_2$

$\dot{x}_2 = -x_1 - x_1^2 x_2.$

THANK YOU

Advanced Control Systems

Module V

Part II

Lyapunov Stability

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Syllabus

5	**Phase Plane and Lyapunov Stability Analysis**	**(8 hours)**
5.1	Phase plots: Concepts- Singular points - Classification of singular points.	1
5.2	Construction of phase trajectories using Isocline method for linear and nonlinear systems	2
5.3	Definition of stability- asymptotic stability and instability	1
5.4	Lyapunov stability analysis: Lyapunov function- Lyapunov methods to stability of nonlinear systems	2
5.5	Lyapunov methods to LTI continuous time systems.	2

Lyapunov Stability

Consider a dynamical system which satisfies

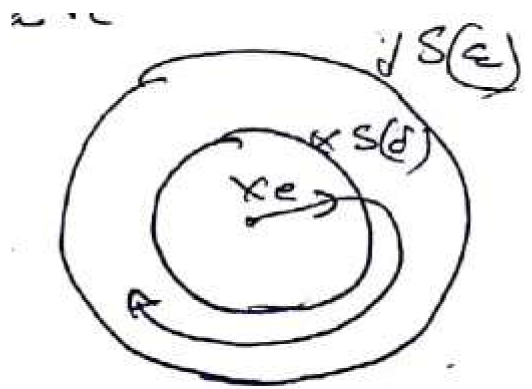
$$\dot{x} = f(x,t) \qquad x(t_0) = x_0 \qquad x \in \mathbb{R}^n. \qquad (4.31)$$

We will assume that f (x, t) satisfies the standard conditions for the existence and uniqueness of solutions. Such conditions are, for instance, that f(x, t) is continuous with respect to x, uniformly in t, and piece-wise continuous in t. A point $x*$ ∈ R_n is an equilibrium point of (4.31) if $f(x*, t) \equiv 0$.

Stability

Definition 4.1. Stability in the sense of Lyapunov
The equilibrium point $x^* = 0$ of (4.31) is *stable (in the sense of Lyapunov)* at $t = t_0$ if for any $\epsilon > 0$ there exists a $\delta(t_0, \epsilon) > 0$ such that

$$\|x(t_0)\| < \delta \quad \Longrightarrow \quad \|x(t)\| < \epsilon, \quad \forall t \geq t_0. \tag{4.32}$$

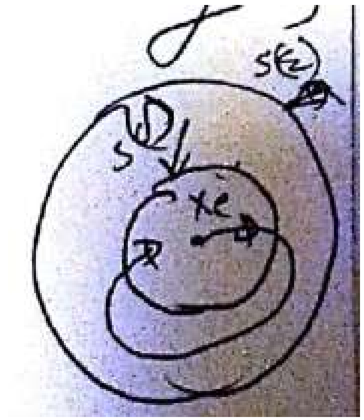


Asymptotic Stability

Definition 4.2. Asymptotic stability
An equilibrium point $x^* = 0$ of (4.31) is *asymptotically stable* at $t = t_0$ if

1. $x^* = 0$ is stable, and
2. $x^* = 0$ is locally attractive; i.e., there exists $\delta(t_0)$ such that

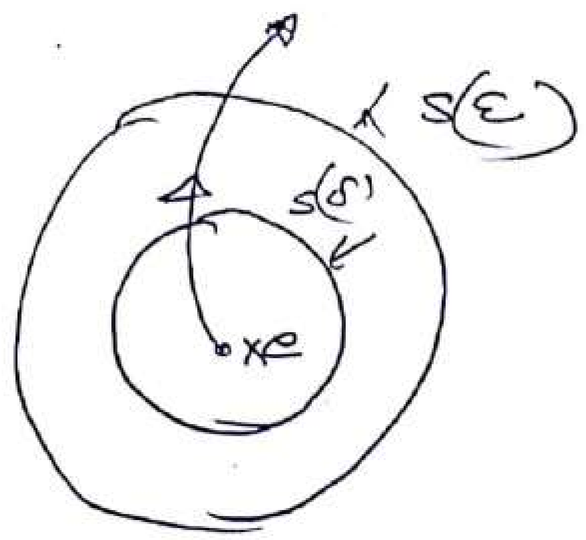
$$\|x(t_0)\| < \delta \implies \lim_{t \to \infty} x(t) = 0. \tag{4.33}$$



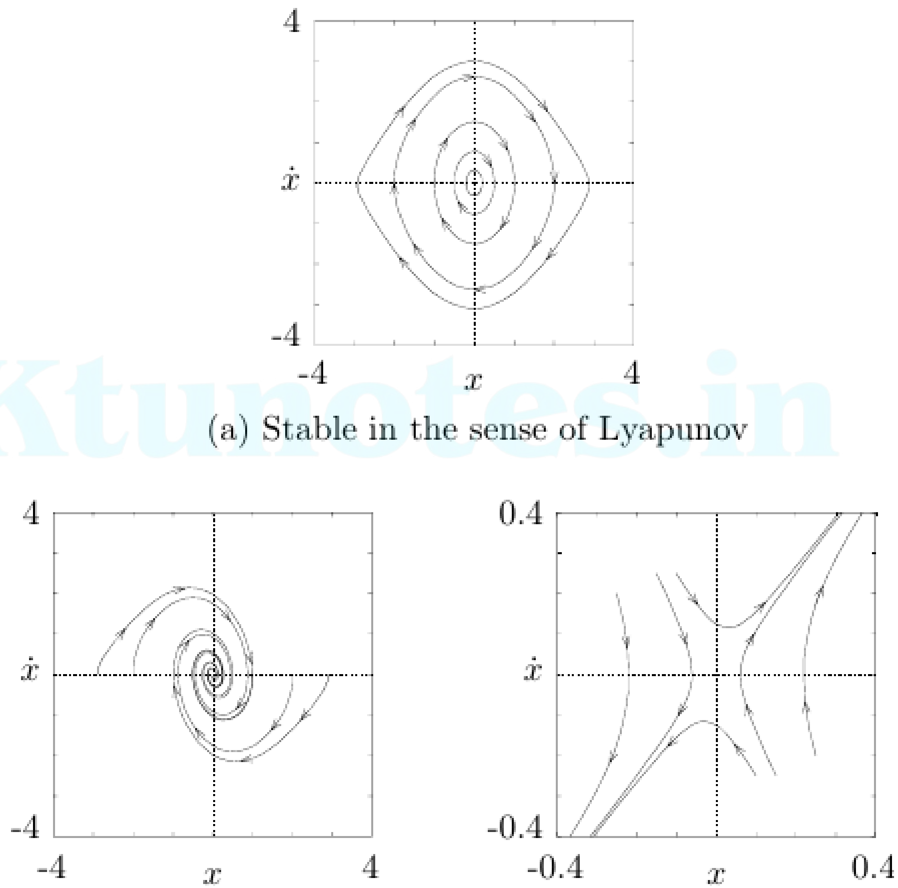
Instability

The equilibrium point x* = 0 of (4.31) is stable (in the sense of Lyapunov) at t = t0 if for any > 0 there exists a δ(t0 ,) > 0 such that

$$\|x(t_0)\| < \delta \quad \Longrightarrow \quad \|x(t)\| > \epsilon, \quad \forall t \geq t_0.$$



Lyapunov Stability….



(a) Stable in the sense of Lyapunov

(b) Asymptotically stable

(c) Unstable (saddle)

Figure 4.7: Phase portraits for stable and unstable equilibrium

Positive Definite and Negative Definite Functions

Definition 4.4. Locally positive definite functions (lpdf)
A continuous function $V : \mathbb{R}^n \times \mathbb{R}_+ \to \mathbb{R}$ is a *locally positive definite function* if for some $\epsilon > 0$ and some continuous, strictly increasing function $\alpha : \mathbb{R}_+ \to \mathbb{R}$,

$$V(0,t) = 0 \quad \text{and} \quad V(x,t) \geq \alpha(\|x\|) \qquad \forall x \in B_\epsilon, \forall t \geq 0. \tag{4.35}$$

A locally positive definite function is locally like an energy function. Functions which are globally like energy functions are called positive definite functions:

Definition 4.5. Positive definite functions (pdf)
A continuous function $V : \mathbb{R}^n \times \mathbb{R}_+ \to \mathbb{R}$ is a *positive definite function* if it satisfies the conditions of Definition 4.4 and, additionally, $\alpha(p) \to \infty$ as $p \to \infty$.

To bound the energy function from above, we define decrescence as follows:

Positive Definite and Negative Definite Functions

- quadratic form

$$V(x) = x'PX$$

$$= [x_1 \; x_2 \; \ldots \; x_n] \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ \vdots & & & \vdots \\ P_{1n} & \cdots & & P_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

positive definites of the quadratic form $V(x)$ can be determined by sylvester's criterion which states that the necessary and sufficient condition that the quadratic form $V(x)$ be positive definite are, that all the successive principal minors of P be positive.

i.e. $P_{11} > 0$; $\begin{vmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{vmatrix} > 0$ . . . . $\begin{vmatrix} P_{11} & \cdots & P_{1n} \\ P_{12} & \cdots & P_{2n} \\ \vdots & & \\ P_{1n} & \cdots & P_{nn} \end{vmatrix} > 0$

The direct method of Lyapunov

Lyapunov's direct method (also called the second method of Lyapunov) allows us to determine the stability of a system without explicitly integrating the differential equation (4.31).

The direct method of Lyapunov

Theorem 4.4. Basic theorem of Lyapunov

Let $V(x,t)$ be a non-negative function with derivative $\dot{V}$ along the trajectories of the system.

1. *If $V(x,t)$ is locally positive definite and $\dot{V}(x,t) \leq 0$ locally in x and for all t, then the origin of the system is locally stable (in the sense of Lyapunov).*

2. *If $V(x,t)$ is locally positive definite and decrescent, and $\dot{V}(x,t) \leq 0$ locally in x and for all t, then the origin of the system is uniformly locally stable (in the sense of Lyapunov).*

3. *If $V(x,t)$ is locally positive definite and decrescent, and $-\dot{V}(x,t)$ is locally positive definite, then the origin of the system is uniformly locally asymptotically stable.*

4. *If $V(x,t)$ is positive definite and decrescent, and $-\dot{V}(x,t)$ is positive definite, then the origin of the system is globally uniformly asymptotically stable.*

- g

Liapunov main stability theorem

Suppose that a system is described by $\dot{x} = f(x,t)$ where $f(0,t) = 0$ for all t. If there exists a scalar function $V(x,t)$ having continuous, first partial derivatives and satisfying the following conditions:

1. $V(x,t)$ is positive definite
2. $\dot{V}(x,t)$ is negative definite

then the equilibrium state at the origin is uniformly asymptotically stable.

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<u>Method of Liapunov</u>

Choose a possible Liapunov function

$$V(x) = X^T P X$$

where P is a positive definite real symmetric matrix.

$$\dot{V}(x) = \dot{X}^T P X + X^T P \dot{X}$$

$$= (AX)^T P X + X^T P \dot{X}$$

$$= X^T A^T P X + X^T P A X$$

$$= X^T \left[A^T P + P A\right] X$$

$$= X^T Q X$$

where $Q = -\left[A^T P + P A\right]$ = positive definite.

∴ For the asymptotic stability of the system, it is sufficient that Q be positive definite.

Problem

1

Consider the system described by

$$\dot{x}_1 = x_2 - x_1(x_1^2 + x_2^2)$$
$$\dot{x}_2 = -x_1 - x_2(x_1^2 + x_2^2)$$

Determine the stability

Define a scalar function $V(x)$

$$V(x) = x_1^2 + x_2^2$$ which is positive definite.

$$\dot{V}(x) = 2x_1 \frac{dx_1}{dt} + 2x_2 \frac{dx_2}{dt}$$

$$= 2x_1 \dot{x}_1 + 2x_2 \dot{x}_2$$

$$= 2x_1 \left[x_2 - x_1(x_1^2 + x_2^2)\right] \cancel{+ 2x_2\left[-x_1 - x_2(x_1^2+x_2^2)\right]}$$

$$+ 2x_2 \left[-x_1 - x_2(x_1^2 + x_2^2)\right]$$

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Problem

1

$$= 2x_1\left[x_2 - x_1^3 - x_1x_2^2\right] + 2x_2\left[-x_1 - x_2x_1^2 - x_2^3\right]$$

$$= \cancel{2x_1x_2} - 2x_1^4 - 2x_1^2x_2^2 - \cancel{2x_1x_2} - 2x_2^2x_1^2 - 2x_2^4$$

$$= -2\left[x_1^4 + 2x_1^2x_2^2 + x_2^4\right]$$

$$= -2\left[x_1^2 + x_2^2\right]^2$$ which is

negative definite.

∴ The equilibrium state at the origin of the system is asymptotically stable in the large.

Problem

2

Determine the stability of the system

$$\dot{x} = AX$$

$$A = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$$

Choose $Q = I$.

$$A^T P + PA = -I$$

$$\begin{bmatrix} -1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Note $P_{12} = P_{21}$.

Solving

$$-2P_{11} + 2P_{12} = -1$$

$$-2P_{11} - 5P_{12} + P_{22} = 0$$

$$-4P_{12} - 8P_{22} = -1$$

Problem

2

solving for P_{ij}'s,

$$P = \begin{bmatrix} \frac{23}{60} & -\frac{7}{60} \\ -\frac{7}{60} & \frac{11}{60} \end{bmatrix}$$

using sylvester's test

$$P_{11} = \frac{23}{60} > 0$$

$$\begin{vmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{vmatrix} = \begin{vmatrix} \frac{23}{60} & -\frac{7}{60} \\ -\frac{7}{60} & \frac{11}{60} \end{vmatrix} > 0$$

$\therefore$ P is positive definite

Hence the equilibrium state at the origin is asymptotically stable in the large.

University Questions

- B

What are singular point? Explain the types of singular point.

Determine given quadratic form is positive definite or not

$$V(x) = 10x_1^2 + 4x_2^2 + x_3^2 + 2x_1 x_2 - 2x_2 x_3 - 4x_1 x_3$$

(10)

A second order system is represented by $\dot{x} = Ax$ where

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Assuming matrix Q to be identity matrix, solve for matrix P in the equation $A^T P + PA = -Q$. Use Lyapunov theorem and determine the stability of the system. Write the Lyapunov function $V(x)$

University Questions

- 2

Define Singular point. Explain the nature of Eigen values of system matrix for any five types of singular points.

Explain Liapunov second method of stability for nonlinear systems.

A linear second order system is described by the equation

$\ddot{e} + 2\delta\omega_n\dot{e} + \omega_n^2 e = 0$

Where $\delta = 0.15$, $\omega_n = 1$rad/sec, e(0)=1.5, and $\dot{e}(0) = 0$

Determine the singular point and state the stability by constructing the phase trajectory using the method of isoclines.

University Questions

- 3 Explain with neat diagram, what is phase trajectory and phase portrait?
  Define positive definite and positive semi definite functions according to Liapunov stability criteria, with suitable examples.

THANK YOU